

MATTHEW J. HEERING

Millstone, NJ | (908) 577-4754 | www.linkedin.com/in/MHeering/ | MJH449@cornell.edu

EDUCATION

Cornell University, College of Engineering, Ithaca, NY
Bachelor of Science, Mechanical Engineering
GPA: 3.79

Expected May 2027

SPECIALIZED SKILLS

Design: SolidWorks, Autodesk Fusion 360, Onshape, DFM, Ansys, FEA

Programming: MATLAB, Python, Arduino

Manufacturing & Prototyping: Manual mill and lathe machining, welding, 3D printing, mechanical assembly, VFD wiring

Relevant Courses: Legged Robots, Fluid Mechanics, Simulations, System Dynamics, Mechanics of Engineering Materials, Dynamics, Statics and Mechanics of Solids, Thermodynamics, Waves and Quantum Physics, Mechanical Design, Computing

ENGINEERING EXPERIENCE

Beanstalk Farms, Manassas, VA

May 2026 – Aug 2026

Mechanical Engineering Intern

- Co-fabricated an 8' × 3' Terra unplugging machine in a 5-week build, cutting and welding steel, assembling hardware, wiring VFDs, and addressing shaft deflection with a support bearing before full-pallet production testing
- Owned harvester conveyor subsystem design in Onshape, applying DFM principles and COTS components
- Prototyped a passive basin/lid separation and flipping mechanism to combine handling operations for automation
- Designed a spinach guide and performed a waste analysis which identified \$1.6k/day in potential product loss; used this ROI analysis to justify project implementation

CSalt, Ithaca, NY

Aug 2024 – Present

Mechanical Subteam Member

- Conceptualized a coral reef regeneration system; now leading the energy generation and stabilization subsystem, developing wave and solar power architecture to provide stable power to an electrochemical cage
- Modeled a 29m × 19m wave energy converter flap supplying energy to pump 0.011 m³/min of water, using Fusion 360
- Co-developed design awarded Best Technical Design in the U.S. DOE 2025 Marine Energy Collegiate Competition

Cornell University Solar Boat Project Team, Ithaca, NY

Aug 2024 – Oct 2025

Drivetrain and Steering Technical Lead & Member

- Directed 11 members in the design, manufacturing, and assembly of the 237-component drivetrain system
- Analyzed propeller performance in MATLAB, determined a 0.83 pitch ratio informing sprint propeller selection
- Evaluated commercial propellers from 4 local marinas; compared geometry, cost, and performance for 7-hp drivetrain

PROJECTS

Robot Felix, Ithaca, NY

April 2026 – May 2026

Engineering Project

- Built and integrated an autonomous cube-collecting robot with motors, H-bridges, an Arduino, and sensors for autonomous cube collection resulting in a 5–1 round-robin record and advancement to the elimination round
- Programmed Arduino in C; implemented timer interrupts, pin-change interrupts, sensor averaging, and threshold logic to process TCS3200 and QTI sensors for autonomous motor control

Wind Turbine Analysis, Ithaca, NY

Mar 2026 – Mar 2026

Coursework

- Performed static structural and modal analyses in Ansys on a wind blade, evaluating stress and natural frequencies against expected excitation frequencies to assess resonance risk

Trash Drone, Millstone, NJ

May 2025 – Present

Independent Engineering Project

- Developed a quadcopter for plastic collection; integrated a servo-powered sweeping arm and flight hardware
- Modeled and 3D-printed a PETG frame in Fusion 360, incorporating mounting geometry for propulsion electronics
- Selected and sized motors, ESCs, a battery, and flight electronics for approximately 2.4 kg of total thrust

ADDITIONAL EXPERIENCE

Sunset Farmstead, Mansfield, NJ

Jun 2023 – Aug 2025

Agricultural Assistant

- Built a new above-ground piping and sprinkler system to automate the watering process and connect plots of crops
- Collaborated with co-workers to pot 72,000 plants per summer, building an ecosystem for bees and monarch butterflies
- Operated heavy machinery and tractors to prepare new batches of soil and streamline the transportation of materials